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UNIVERSITAT RAMON LLULL

Databases project 2025-2026
Choose Your Adventure

Group 5

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Table of Contents

01 Project Overview

02 Brief Dataset Description

03 Data Model Design

04 Key Design Decisions

05 Physical Implementation

06 Query Strategy

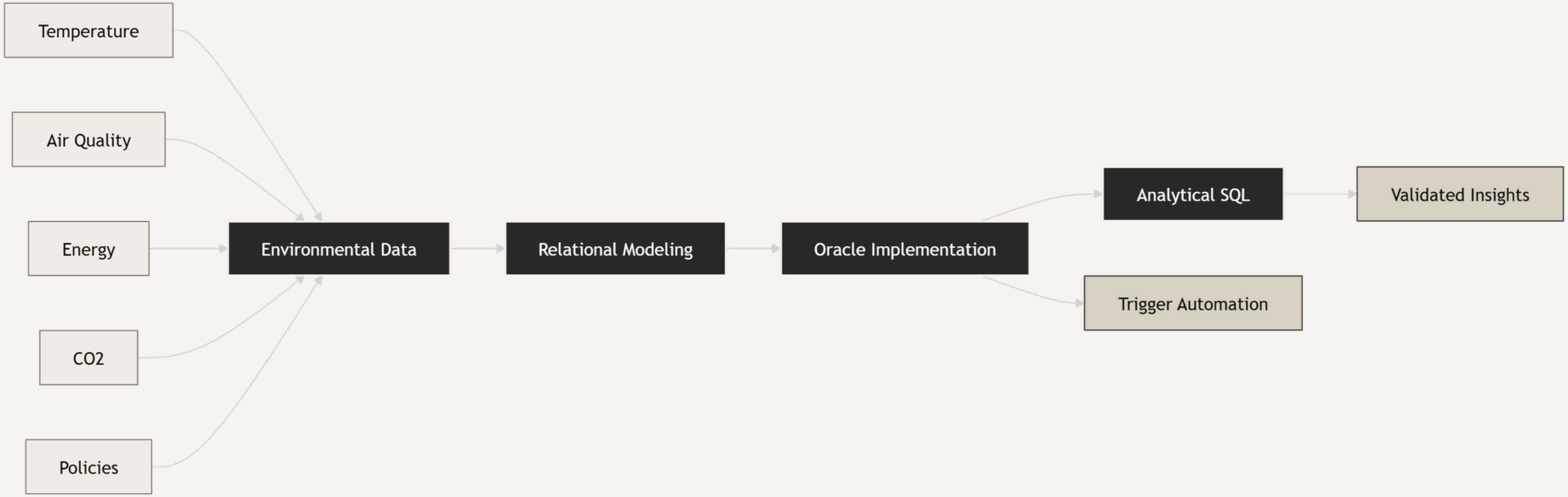
07 Query Examples

08 Result Analysis

09 Limitations & Problems

10 Conclusions & References

Project Overview



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Table of Contents

01 Project Overview

02 Brief Dataset Description

03 Data Model Design

04 Key Design Decisions

05 Physical Implementation

06 Query Strategy

07 Query Examples

08 Result Analysis

09 Limitations & Problems

10 Conclusions & References

Brief Dataset Description

Environment.zip



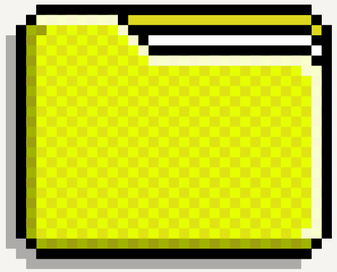
Raw Data

- city_temperature
- co-emissions-by-sector
- Cost_of_living_index
- energy
- global_air_quality_data_10000
- nuclear_power_dataset



Our Own Dataset

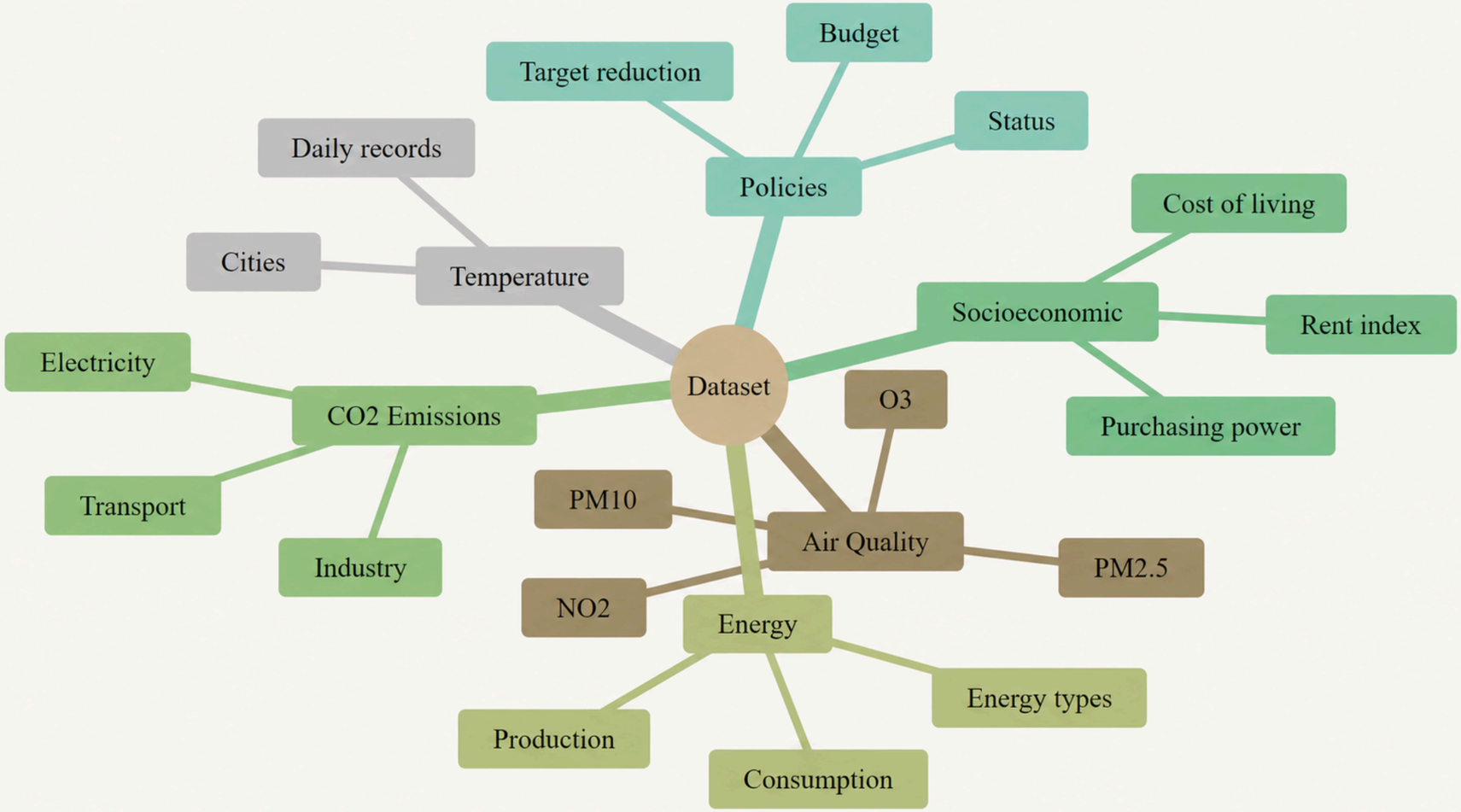
- Government_Policy



Government_Policy.csv:

```
policy_id,country_name,start_policy_year,policy_name,policy_type,budget_million_usd,target_reduction_percent,policy_statu
1,Netherlands,2015,Nuclear Phase-Out,Nuclear Regulation,1466,14,Active
2,Sao Tome and Principe,2001,Emissions Trading Scheme,Nuclear Regulation,4058,77,Cancelled
```

Brief Dataset Description



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Table of Contents

01 Project Overview

02 Brief Dataset Description

03 Data Model Design

04 Key Design Decisions

05 Physical Implementation

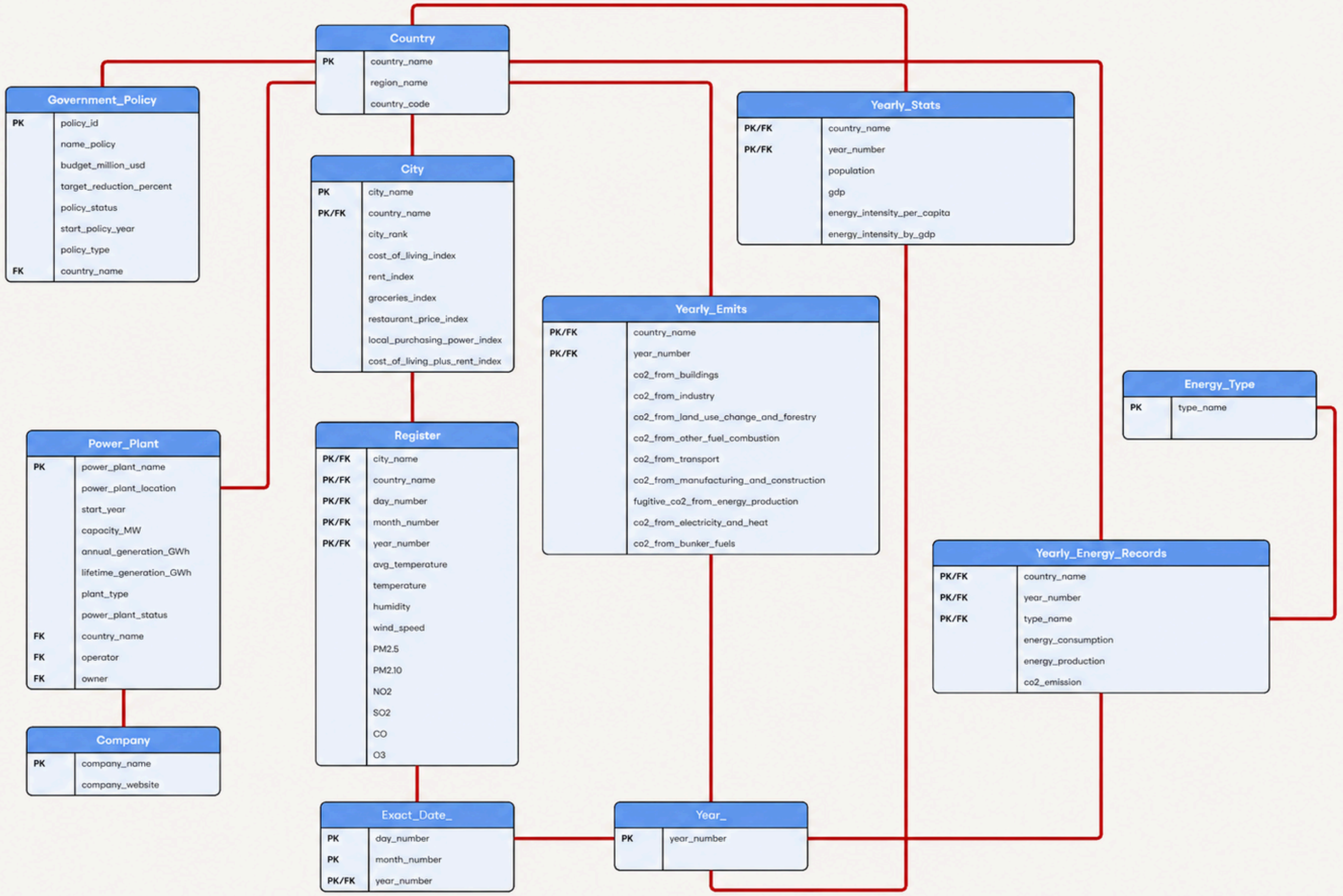
06 Query Strategy

07 Query Examples

08 Result Analysis

09 Limitations & Problems

10 Conclusions & References


Country
PK country_name
region_name
country_code
Government_Policy
PK policy_id
name_policy
budget_million_usd
target_reduction_percent
policy_status
start_policy_year
policy_type
FK country_name
City
PK city_name
PK/FK country_name
city_rank
cost_of_living_index
rent_index
groceries_index
restaurant_price_index
local_purchasing_power_index
cost_of_living_plus_rent_index
Yearly_Stats
PK/FK country_name
PK/FK year_number
population
gdp
energy_intensity_per_capita
energy_intensity_by_gdp
Yearly_Emits
PK/FK country_name
PK/FK year_number
co2_from_buildings
co2_from_industry
co2_from_land_use_change_and_forestry
co2_from_other_fuel_combustion
co2_from_transport
co2_from_manufacturing_and_construction
fugitive_co2_from_energy_production
co2_from_electricity_and_heat
co2_from_bunker_fuels
Energy_Type
PK type_name
Power_Plant
PK power_plant_name
power_plant_location
start_year
capacity_MW
annual_generation_GWh
lifetime_generation_GWh
plant_type
power_plant_status
FK country_name
FK operator
FK owner
Register
PK/FK city_name
PK/FK country_name
PK/FK day_number
PK/FK month_number
PK/FK year_number
avg_temperature
temperature
humidity
wind_speed
PM2.5
PM2.10
NO2
SO2
CO
O3
Yearly_Energy_Records
PK/FK country_name
PK/FK year_number
PK/FK type_name
energy_consumption
energy_production
co2_emission
Company
PK company_name
company_website
Exact_Date_
PK day_number
PK month_number
PK/FK year_number
Year_
PK year_number

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Table of Contents

01 Project Overview

02 Brief Dataset Description

03 Data Model Design

04 Key Design Decisions

05 Physical Implementation

06 Query Strategy

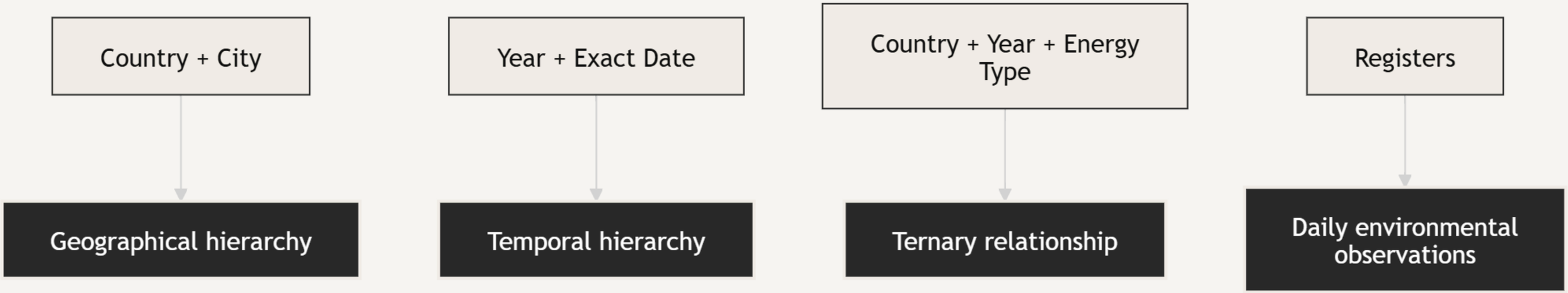
07 Query Examples

08 Result Analysis

09 Limitations & Problems

10 Conclusions & References

Key Decisions



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Table of Contents

01 Project Overview

02 Brief Dataset Description

03 Data Model Design

04 Key Design Decisions

05 Physical Implementation

06 Query Strategy

07 Query Examples

08 Result Analysis

09 Limitations & Problems

10 Conclusions & References

Physical Implementation

Primary and foreign keys were added to preserve data integrity.



The conceptual and relational designs were translated into SQL tables.

Each entity became a physical table with selected data types.

The database was ready to support queries, validations and triggers.

Table of Contents

01 Project Overview

02 Brief Dataset Description

03 Data Model Design

04 Key Design Decisions

05 Physical Implementation

06 Query Strategy

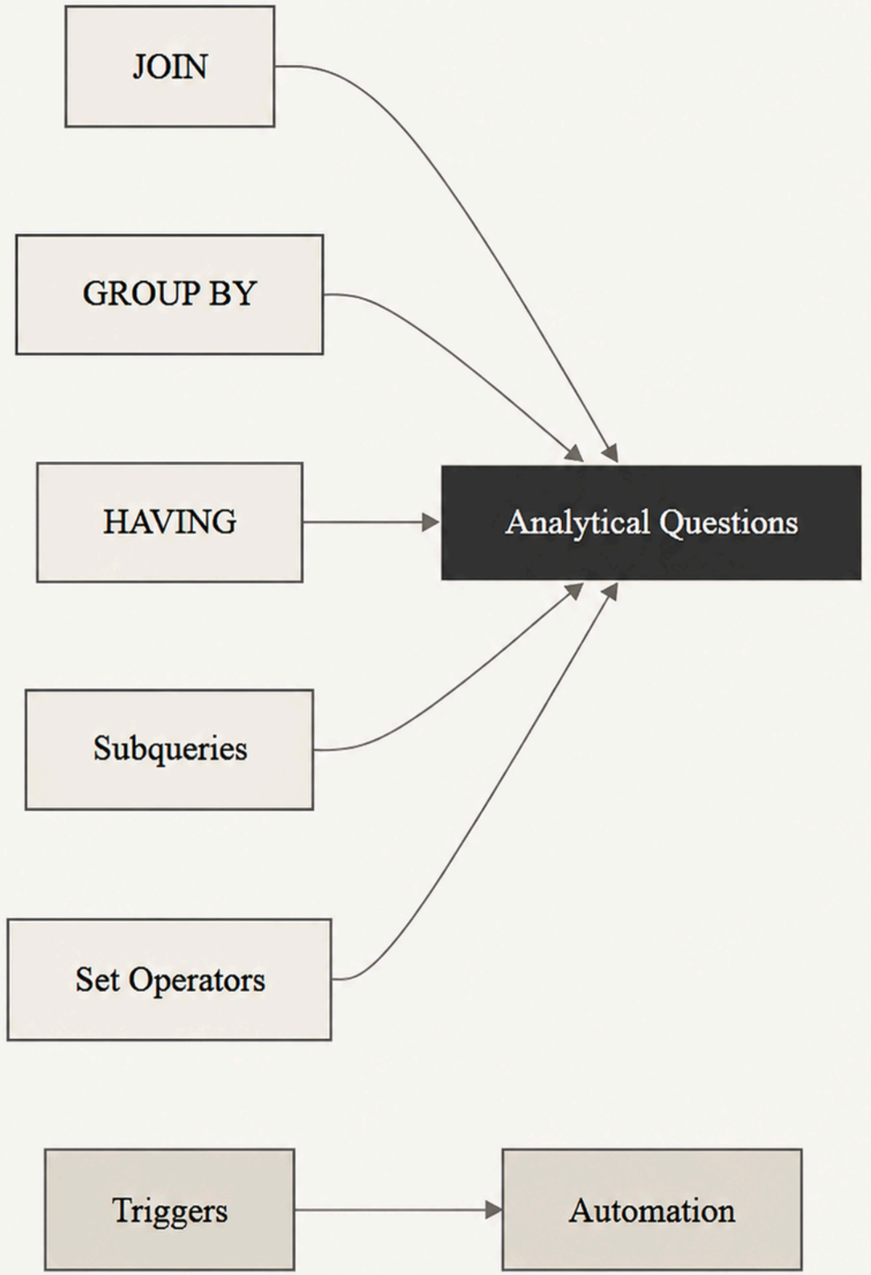
07 Query Examples

08 Result Analysis

09 Limitations & Problems

10 Conclusions & References

Query Strategy



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Table of Contents

01 Project Overview

02 Brief Dataset Description

03 Data Model Design

04 Key Design Decisions

05 Physical Implementation

06 Query Strategy

07 Query Examples

08 Result Analysis

09 Limitations & Problems

10 Conclusions & References

Which countries were both top CO_2 emitters and had active emissions reduction policies in 2015?

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```sql
SELECT yer.country_name AS "Country Name"
FROM Yearly_Energy_Records yer
WHERE yer.year_number = 2015 AND yer.type_name = 'all_energy_types' AND yer.country_name !=
'World' AND yer.co2_emission > (
    SELECT AVG(yer2.co2_emission)
    FROM Yearly_Energy_Records yer2
    WHERE yer2.year_number = 2015 AND yer2.type_name = 'all_energy_types' AND yer2.country_name
!= 'World')
INTERSECT
SELECT gp.country_name AS "Country Name"
FROM Government_Policy gp
WHERE gp.policy_status = 'Active' AND gp.policy_type = 'Emissions Reduction'
ORDER BY "Country Name";
```

Which countries have increased their energy consumption every year between 2010 and 2020?

```sql
SELECT actual.country_name AS "Country Name"
FROM Yearly_Energy_Records actual
LEFT JOIN Yearly_Energy_Records previo ON actual.country_name = previo.country_name
  AND actual.type_name = previo.type_name
  AND actual.year_number = previo.year_number + 1
WHERE actual.type_name = 'all_energy_types' AND actual.year_number BETWEEN 2010 AND 2019
GROUP BY actual.country_name
HAVING COUNT(actual.year_number) = 10 AND SUM(
       CASE
           WHEN previo.energy_consumption IS NOT NULL AND actual.energy_consumption <=
previo.energy_consumption THEN 1
           ELSE 0
       END
   ) = 0
ORDER BY "Country Name" ASC;
```

Resultado de la Consulta | Salida

Descargar

	COUNTRY NAME
1	Bangladesh
2	Cambodia
3	Costa Rica
4	Ethiopia
5	Former Czechoslo...
6	Former Serbia and...
7	Former U.S.S.R.
8	Former Yugoslavia
9	Germany, East
10	Germany, West
11	Guatemala
12	Hawaiian Trade Zo...
13	India
14	Laos
15	Mali
16	Mauritius
17	Peru
18	Rwanda
19	World

Which countries appear as extreme cases in 2015, either very high GDP or very high CO_2 emissions?

```sql
SELECT ys.country_name AS "Country Name", 'High GDP' AS "Category", ys.gdp AS "Value"
FROM Yearly_Stats ys
WHERE ys.year_number = 2015 AND ys.country_name != 'World' AND ys.gdp > (
    SELECT AVG(ys2.gdp)
    FROM Yearly_Stats ys2
    WHERE ys2.year_number = 2015 AND ys2.country_name != 'World')
UNION ALL
SELECT yer.country_name AS "Country Name", 'High CO2 Emission' AS "Category", yer.co2_emission
AS "Value"
FROM Yearly_Energy_Records yer
WHERE yer.year_number = 2015 AND yer.type_name = 'all_energy_types' AND yer.country_name !=
'World' AND yer.co2_emission > (
    SELECT AVG(yer2.co2_emission)
    FROM Yearly_Energy_Records yer2
    WHERE yer2.year_number = 2015 AND yer2.type_name = 'all_energy_types' AND yer2.country_name
!= 'World')
ORDER BY "Country Name", "Category";
```

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Table of Contents

01 Project Overview

02 Brief Dataset Description

03 Data Model Design

04 Key Design Decisions

05 Physical Implementation

06 Query Strategy

07 Query Examples

08 Result Analysis

09 Limitations & Problems

10 Conclusions & References

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Result Analysis



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Table of Contents

01 Project Overview

02 Brief Dataset Description

03 Data Model Design

04 Key Design Decisions

05 Physical Implementation

06 Query Strategy

07 Query Examples

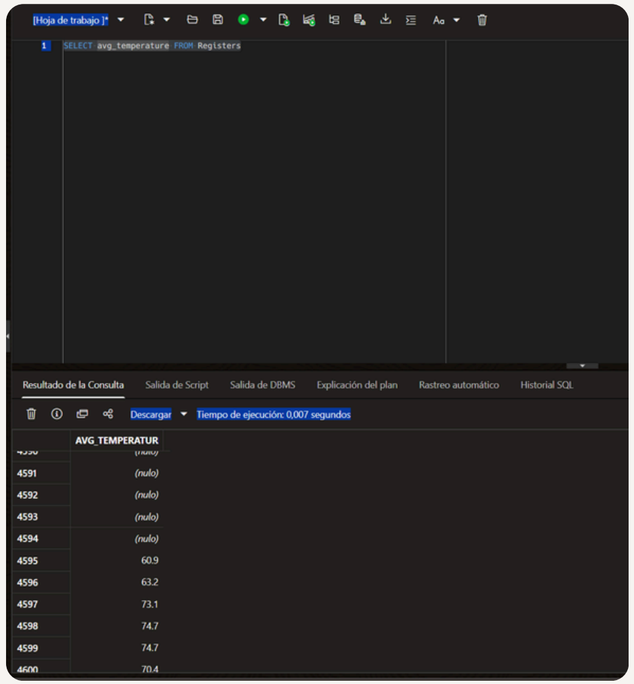
08 Result Analysis

09 Limitations & Problems

10 Conclusions & References

Limitations and Problems

01



Dataset Limitations

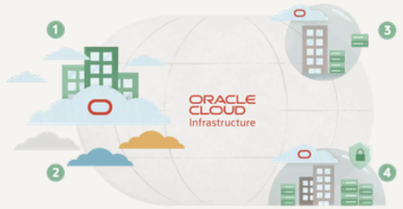
At first instance:

- Lack of dataset information
- Wrong CSV importation(inserts)



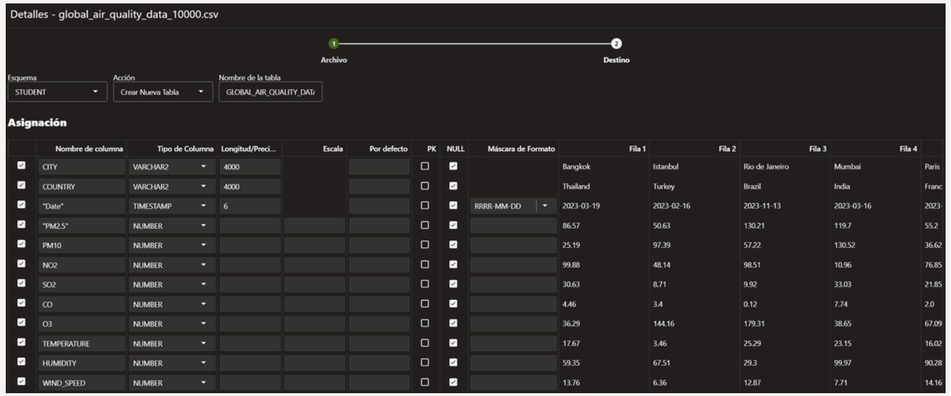
Limitation: in *city_temperature.csv* it is registered but only between [1995, 2020]

02



Oracle Syntax

Handling real-world CSV anomalies, special characters in headers (by default, Oracle states: "PM2.5"), and complex date parsing errors (ORA-01858).



- ”PM2.5” ✗
- PM2_5 ✓

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Table of Contents

01 Project Overview

02 Brief Dataset Description

03 Data Model Design

04 Key Design Decisions

05 Physical Implementation

06 Query Strategy

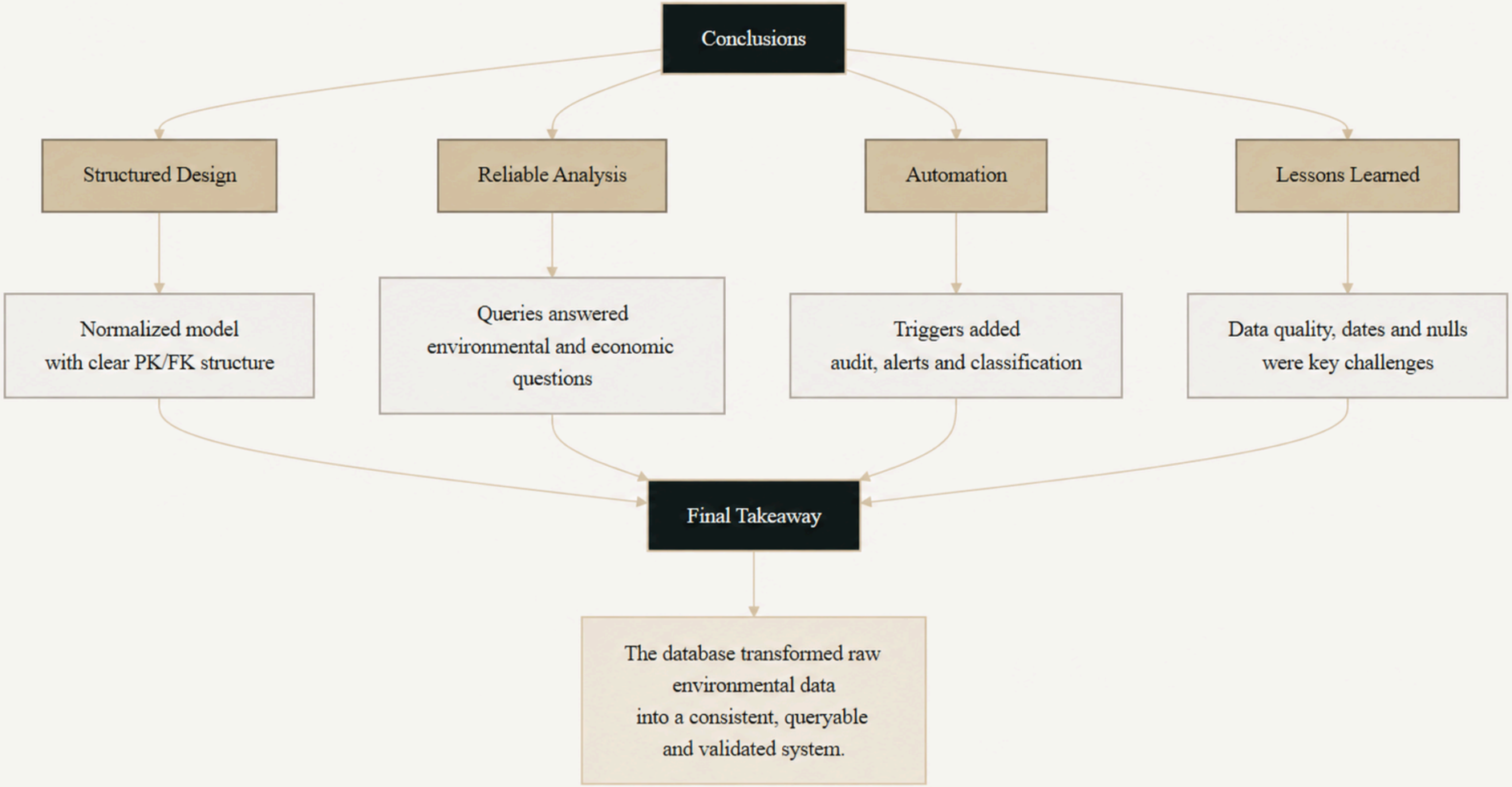
07 Query Examples

08 Result Analysis

09 Limitations & Problems

10 Conclusions & References

Conclusions



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Thank You

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